



Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

A-level COMPUTER SCIENCE

Potential Section C Questions

November 2025

Time allowed: 20 minutes

Materials

For this paper you must have:

- A copy of the Skeleton Program
- A copy of the Preliminary Material

Instructions

- Answer **all** questions.
- Show all rough work for this paper.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 14.

For Examiner's Use	
Question	Mark
1	
TOTAL	



0 6

This refers to the `ChooseCellToMoveTo` method in the `WorkerAnt` class.

0 6. 0

A `WorkerAnt` is **not** carrying food. Describe the process it uses to decide which cell to move to next.

[3 marks]

3

0 7. 1

In the `Simulation` class, the `GetNestInCell` method iterates through the `Nests` collection to find a nest.

State the time complexity for this approach.

[1 marks]

0 7. 2

The `AdvanceStage` method within the `Simulation` class iterates through the `Pheromones` collection.

Explain why a separate temporary list is created to store pheromones that need to be removed, rather than removing them directly inside the main loop.

[2 marks]

3

0 8. 1

State the identifier for a class which inherits from the `Ant` class.

[1 marks]

0 8. 2

State the identifier of a variable local to the `Simulation` class.

[1 marks]

0 8. 3

State the identifier of a method within the `Simulation` class that takes two integers as its parameters and returns a string.

[1 marks]

0 8. 4

State the coordinate of the only guaranteed `Nest` in any simulation.

[1 marks]

0 8. 5

The method `GetAreaDetails` uses string concatenation.

Define the term 'string concatenation'.

[1 marks]

5



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Another alternative to storing the Grid as a linear list would be to use a two dimensional array.

Explain why this may be a more appropriate technique.

[2 marks]

2

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The Cell class inherits from Entity. The Cell constructor includes a call to the constructor of the parent class.

Explain the purpose of calling the parent class constructor.

[1 marks]

1

